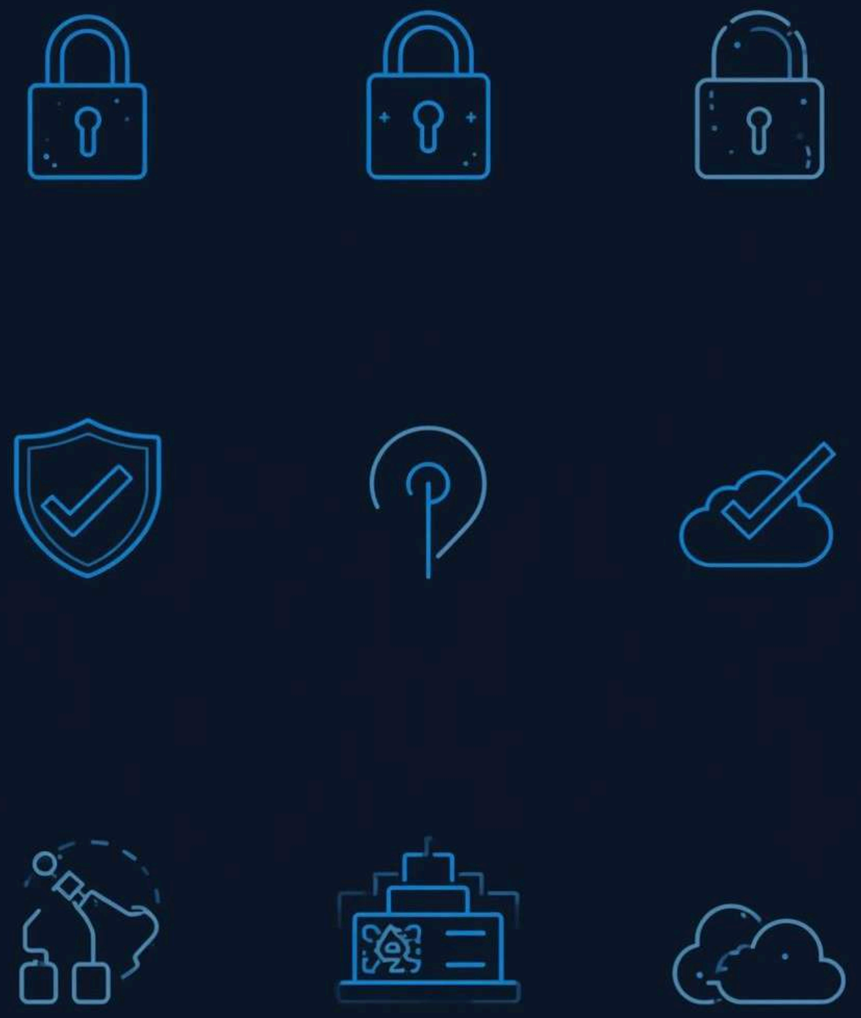
Welcome to Cyber Security

What You Will Learn

Key topics in cyber security for beginners to understand

In this module, we will explore concepts of cyber security. You will learn about the **CIA triad**, discover the various types of **hackers and attacks**, and understand effective **protection methods**. Real-life examples will help you see the importance of cyber security in everyday life.



The CIA Triad

Understanding the Core Principles of Cyber Security

Key Principles

- Confidentiality
- Integrity
- Availability

Importance of Each

- Data protection
- System reliability
- User trust

Confidentiality

Ensuring only authorized access to sensitive information is important.

Confidentiality is about protecting sensitive information. It ensures that only authorized people can access private data.

- Only allowed people can access the data
- Uses passwords and encryption
- Keeps private data secret
- Example: Your email password

Integrity

Understanding the importance of data accuracy and protection methods

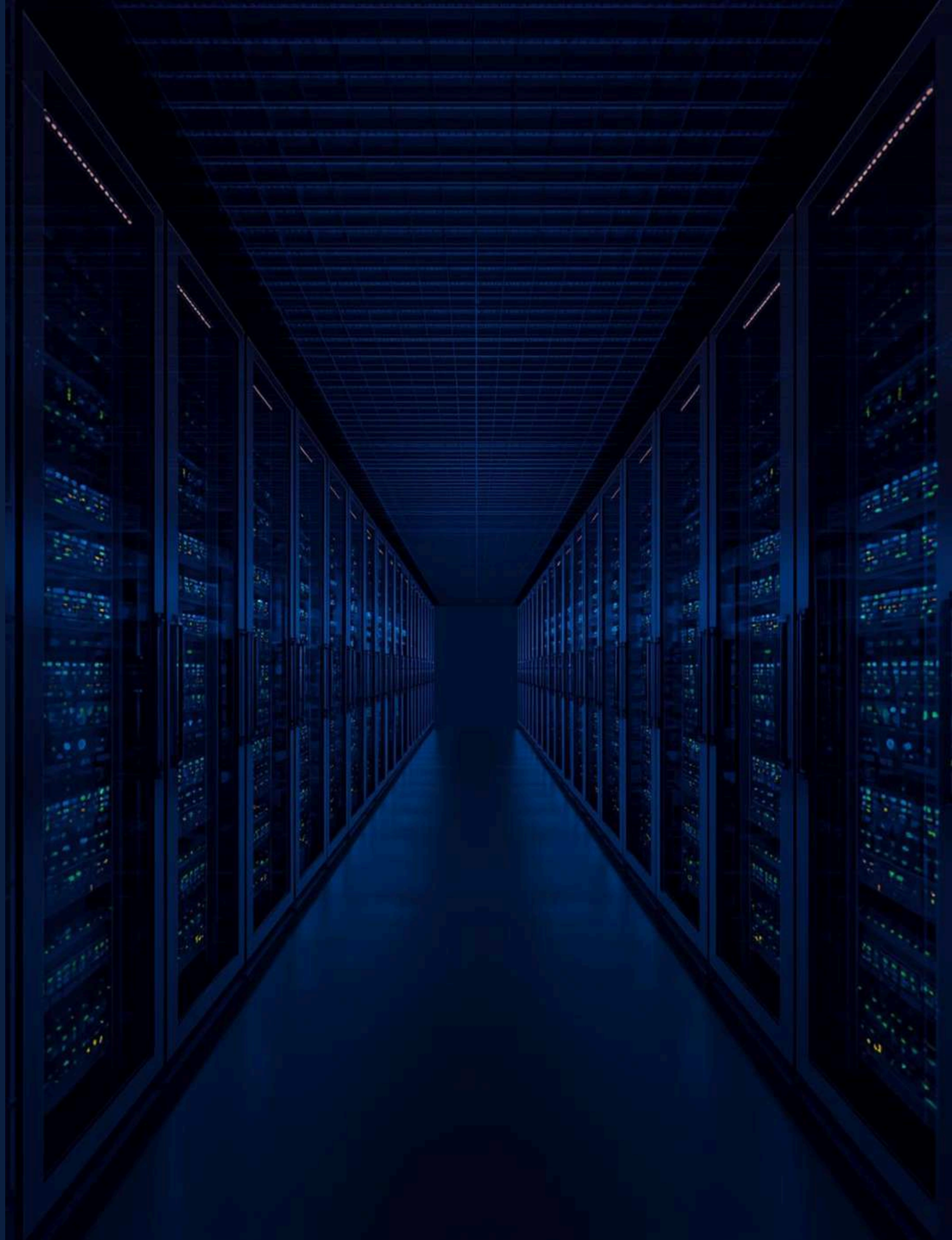
- Data must stay correct and safe
- No one should change it without permission
- Uses checks and protection tools
- Example: Exam marks in the system

Availability

Ensuring systems are always accessible and operational for users

Availability is important in cyber security. It means that systems and data must be accessible whenever needed. Systems must be ready to use anytime

- Uses backups and servers
- Prevents downtime or system failure
- Example: Hospital system must always work



Who Are Hackers?

Understanding the different types and their motivations in cyber security

Hackers are individuals who use their skills to break into systems. They can have various motivations, from ethical to bad. Understanding these differences is important, as not all hackers are bad. Many work to protect systems and improve security measures, helping to create a safer digital world.

Types of Cyber Attacks

Understanding common threats to keep your data safe online

Cyber attacks are ways that hackers use to break into systems and steal information.
Common types include:

- Phishing: tricking people into sharing personal information
- Malware: harmful software that damages or slows down systems
- Denial of Service (DoS): flooding a system to make it stop working
- Hacking: getting into devices or networks without permission

Protecting Our Data

important methods to safeguard information in the digital world

In today's connected world, keeping our data safe is very important.
We can stay safer by using antivirus software, firewalls, and strong passwords.
Updating our systems regularly helps fix security weaknesses.
By following these steps, we can protect our information and use the internet with more confidence.

Real Life Examples

How Cyber Security Protects Our Daily Lives

In our everyday life, cyber security plays a important role in safeguarding sensitive information. From online banking transactions to securing emails and protecting school systems, strong security measures ensure that our data remains private and secure.